Faithfulness analysis (Step 1 and Step 2a)

All values are formulas linked to the "Results" sheet. Grey rows = Gemma (excluded in Step 1). P1 = zero-shot, P2 = few-shot.

0 SETTINGS & SUMMARY

Settings (change here – all tables update)

Setting	Value
Gate threshold – weighted faithfulness	0.75

Weighted score = 0.15-fact identification + 0.30-fact fidelity + 0.15-foil identification + 0.30-foil fidelity + 0.10-system-state correctness

Legend

P1 / P2	zero-shot / few-shot prompting
C1 / C2 / C3	configurations 1–3
TC1 / TC2 / TC3	test cases 1–3 (TC1, TC2 = CC1 · TC3 = CC3)
Hallucination	invented or contradicting content compared to rules.yaml / system snapshot. Omissions and plain-language paraphrases (e.g. "room empty") are not counted.
Yes rate	share of explanations flagged as hallucination (yes / all checked)

Key numbers

Measure	P1 (zero-shot)	P2 (few-shot)
Mean faithfulness (weighted)	0.725	0.725
Combinations passing weighted gate (of 9, Gemma exclu	6	3
Hallucination yes rate	32.4%	42.6%
Hallucinated explanations passing weighted gate	11.4%	19.6%
Median duration (ms)	3,541	5,107
Mean words	48.9	58.6

Sections: 1 Faithfulness · 2 Practicality · 3 Hallucinations · 4 Does the score catch hallucinations?

1 FAITHFULNESS

1.1 Mean faithfulness by model

Model	Weighted P1	Weighted P2	Weighted all
deepseek-v4-flash-0731	0.734	0.718	0.726
openai-gpt-oss-120b	0.776	0.778	0.777
gemma-4-31b-it	*0.647*	*0.713*	*0.680*
mistral-medium-3.5-128b	0.742	0.693	0.717

1.2 Mean faithfulness by prompting style

Prompting style	Weighted	n
P1	0.725	108
P2	0.725	108

1.3 Mean faithfulness by configuration

Configuration	Weighted P1	Weighted P2	Weighted all
C1	0.728	0.713	0.721
C2	0.744	0.746	0.745
C3	0.701	0.717	0.709

1.4 Gate check: model × configuration (P1 vs P2 side by side)

Pass = mean ≥ threshold from Settings. "Better" = higher mean; "equal" if identical.

Model	Config	Weighted P1	Weighted P2	Better (weighted)	Pass weighted P1	Pass weighted P2
deepseek-v4-flash-0731	C1	0.739	0.750	P2	no	yes

deepseek-v4-flash-0731	C2	0.772	0.700	P1	yes	no
deepseek-v4-flash-0731	C3	0.692	0.703	P2	no	no
openai-gpt-oss-120b	C1	0.750	0.772	P2	yes	yes
openai-gpt-oss-120b	C2	0.789	0.822	P2	yes	yes
openai-gpt-oss-120b	C3	0.789	0.739	P1	yes	no
gemma-4-31b-it	*C1*	*0.650*	*0.644*	*P1*	*excluded (Step 1)*	*excluded (Step 1)*
gemma-4-31b-it	*C2*	*0.650*	*0.728*	*P2*	*excluded (Step 1)*	*excluded (Step 1)*
gemma-4-31b-it	*C3*	*0.642*	*0.767*	*P2*	*excluded (Step 1)*	*excluded (Step 1)*
mistral-medium-3.5-128b	C1	0.775	0.686	P1	yes	no
mistral-medium-3.5-128b	C2	0.767	0.733	P1	yes	no
mistral-medium-3.5-128b	C3	0.683	0.658	P1	no	no
Count "yes" (passes gate, Gemma excluded)					**6**	**3**
Count "P1" better				**6**		
Count "P2" better				**6**		
Count "equal"				**0**		

2 PRACTICALITY (speed & length)

Median used for duration because of outliers (e.g. DeepSeek C2 P1). Durations in ms. Words = words in llm_explanation. "≈ equal" = medians differ by < 5 %.

2.1 By prompting style

Prompting style	Median duration (ms)	Mean duration (ms)	Mean words	n
P1	3,541	20,963	48.9	108
P2	5,107	25,618	58.6	108

2.2 By model

Model	Median ms P1	Median ms P2	Faster	Mean words P1	Mean words P2
deepseek-v4-flash-0731	2,098	4,134	P1	60.3	70.1
openai-gpt-oss-120b	6,045	8,669	P1	54.0	55.7
gemma-4-31b-it	*2,850*	*5,249*	*P1*	*36.6*	*54.1*
mistral-medium-3.5-128b	2,871	3,817	P1	44.8	54.3

2.3 By model × configuration

Model	Config	Median ms P1	Median ms P2	Faster
deepseek-v4-flash-0731	C1	1,264	1,736	P1
deepseek-v4-flash-0731	C2	87,962	8,238	P2
deepseek-v4-flash-0731	C3	1,749	3,349	P1
openai-gpt-oss-120b	C1	4,323	4,314	≈ equal
openai-gpt-oss-120b	C2	11,755	18,007	P1
openai-gpt-oss-120b	C3	5,410	6,309	P1
gemma-4-31b-it	*C1*	*1,068*	*1,909*	*P1*
gemma-4-31b-it	*C2*	*5,826*	*6,547*	*P1*
gemma-4-31b-it	*C3*	*2,850*	*4,197*	*P1*
mistral-medium-3.5-128b	C1	1,583	2,329	P1
mistral-medium-3.5-128b	C2	5,514	7,166	P1
mistral-medium-3.5-128b	C3	2,810	3,817	P1
Count "P1" faster				**10**
Count "P2" faster				**1**
Count "≈ equal"				**1**

3 HALLUCINATIONS

Change = yes rate P2 − yes rate P1 (positive = more hallucinations with few-shot).

3.1 By model

Model	yes P1	n P1	Yes rate P1	yes P2	n P2	Yes rate P2	Yes rate all	Change (P2 − P1)
deepseek-v4-flash-0731	8	27	29.6%	13	27	48.1%	38.9%	+18.5%
openai-gpt-oss-120b	10	27	37.0%	7	27	25.9%	31.5%	-11.1%
gemma-4-31b-it	*7*	*27*	*25.9%*	*14*	*27*	*51.9%*	*38.9%*	*+25.9%*
mistral-medium-3.5-128b	10	27	37.0%	12	27	44.4%	40.7%	+7.4%
Total	**35**	**108**	**32.4%**	**46**	**108**	**42.6%**	**37.5%**	**+10.2%**

3.2 By configuration

Configuration	yes P1	n P1	Yes rate P1	yes P2	n P2	Yes rate P2	Yes rate all	Change (P2 − P1)
C1	17	36	47.2%	18	36	50.0%	48.6%	+2.8%
C2	13	36	36.1%	16	36	44.4%	40.3%	+8.3%
C3	5	36	13.9%	12	36	33.3%	23.6%	+19.4%
Total	**35**	**108**	**32.4%**	**46**	**108**	**42.6%**	**37.5%**	**+10.2%**

3.3 By test case

Test case	yes P1	n P1	Yes rate P1	yes P2	n P2	Yes rate P2	Yes rate all	Change (P2 − P1)
TC1	1	36	2.8%	3	36	8.3%	5.6%	+5.6%
TC2	20	36	55.6%	21	36	58.3%	56.9%	+2.8%
TC3	14	36	38.9%	22	36	61.1%	50.0%	+22.2%
Total	**35**	**108**	**32.4%**	**46**	**108**	**42.6%**	**37.5%**	**+10.2%**

3.4 Model × configuration

Counts of "yes" (n = 9 explanations per cell).

Model	C1 P1	C2 P1	C3 P1	Total P1	C1 P2	C2 P2	C3 P2	Total P2
deepseek-v4-flash-0731	5	3	0	8	6	5	2	13
openai-gpt-oss-120b	3	4	3	10	3	3	1	7
gemma-4-31b-it	*3*	*3*	*1*	*7*	*3*	*5*	*6*	*14*
mistral-medium-3.5-128b	6	3	1	10	6	3	3	12

3.5 Model × test case

Counts of "yes" (n = 9 explanations per cell).

Model	TC1 P1	TC2 P1	TC3 P1	Total P1	TC1 P2	TC2 P2	TC3 P2	Total P2
deepseek-v4-flash-0731	0	5	3	8	0	6	7	13
openai-gpt-oss-120b	0	6	4	10	0	4	3	7
gemma-4-31b-it	*1*	*6*	*0*	*7*	*3*	*8*	*3*	*14*
mistral-medium-3.5-128b	0	3	7	10	0	3	9	12

3.6 Hallucination types

Only rows flagged "yes" are counted. Gemma columns in grey.

Type	DeepSeek P1	GPT-OSS P1	Gemma P1	Mistral P1	Total P1	DeepSeek P2	GPT-OSS P2	Gemma P2	Mistral P2	Total P2
invented priority	5	5	*7*	3	20	6	2	*10*	3	21
sensor-state error	3	3	*0*	1	7	6	4	*4*	9	23
invented trigger/condition	0	1	*0*	5	6	0	1	*0*	0	1
misread rule file	0	1	*0*	0	1	0	0	*0*	0	0
speculative malfunction	0	0	*0*	1	1	1	0	*0*	0	1
confusing wording	0	0	*0*	0	0	0	0	*0*	0	0
Total	**8**	**10**	*7*	**10**	**35**	**13**	**7**	*14*	**12**	**46**

3.7 C3 only: did changing the foil reduce hallucinations?

Foil kept = LLM kept the TOPSIS foil (foil_match TRUE).

C3 foil	yes P1	n P1	Yes rate P1	yes P2	n P2	Yes rate P2
foil kept (= TOPSIS)	4	21	19.0%	9	24	37.5%
foil changed by LLM	1	15	6.7%	3	12	25.0%

4 DOES THE FAITHFULNESS SCORE CATCH HALLUCINATIONS?

If the score worked as a hallucination filter, hallucinated explanations should rarely pass the gate.

Group	n	Mean weighted	Pass weighted gate (n)	Pass weighted gate (share)
P1 – hallucinated	35	0.684	4	11.4%
P1 – not hallucinated	73	0.744	36	49.3%
P2 – hallucinated	46	0.683	9	19.6%
P2 – not hallucinated	62	0.756	30	48.4%

Reading: hallucinated explanations rarely pass the weighted gate (P1: 4/35, P2: 9/46), but about half of the explanations without hallucinations do not pass it either (P1: 37/73, P2: 32/62). A low score does not show whether an explanation is wrong or only incomplete, so a separate hallucination check is still needed.

5 PROMPTING STRATEGY: P1 vs P2 ACROSS ALL MODEL × CONFIGURATION PAIRS

Question: which prompting strategy should the system use? The model is an interchangeable component, so the strategy is judged across all pairs, not by its best single cell.
Based on the weighted faithfulness score per model × configuration (table 1.4, columns C and D). "≈ equal" = difference < 0.01.
Minimax view: choose the strategy with the better worst case – it degrades most gracefully if a different model is used later.

5.1 Weighted faithfulness per model × configuration

Measure	P1 – all 12	P2 – all 12	Better (all 12)	P1 – without Gemma (9)	P2 – without Gemma (9)	Better (without Gemma)
Mean (macro-average)	0.725	0.725	**≈ equal**	0.751	0.729	**P1**
Median	0.744	0.731	**P1**	0.767	0.733	**P1**
Worst case (minimum)	0.642	0.644	**≈ equal**	0.683	0.658	**P1**
Best case (maximum)	0.789	0.822	**P2**	0.789	0.822	**P2**
Standard deviation (spread) – lower is better	0.058	0.050	**≈ equal**	0.039	0.049	**≈ equal**
Range (max – min) – lower is better	0.147	0.178	**P1**	0.106	0.164	**P1**
Cells ≥ weighted threshold (B9)	6	4	**P1**	6	3	**P1**
Count "P1" better			**3**			**5**
Count "P2" better			**1**			**1**
Count "≈ equal"			**3**			**1**

Reading: across all 12 pairs, P1 is better on 3 of 7 measures, P2 on 1, and 3 are about equal. Without Gemma (excluded in Step 1), P1 is better on 5 of 7, P2 on 1. The single best cell (P2) does not reflect the strategy-level result.

5.2 Hallucination yes rate per model × configuration (lower is better)

Each cell = share of the 9 explanations flagged "yes". Rates move in steps of 11.1 % (1 of 9). "More hallucinations" compares P1 and P2 within the same cell.

Model	Config	Yes rate P1	Yes rate P2	More hallucinations
deepseek-v4-flash-0731	C1	55.6%	66.7%	P2
deepseek-v4-flash-0731	C2	33.3%	55.6%	P2
deepseek-v4-flash-0731	C3	0.0%	22.2%	P2
openai-gpt-oss-120b	C1	33.3%	33.3%	equal
openai-gpt-oss-120b	C2	44.4%	33.3%	P1
openai-gpt-oss-120b	C3	33.3%	11.1%	P1
gemma-4-31b-it	*C1*	*33.3%*	*33.3%*	*equal*
gemma-4-31b-it	*C2*	*33.3%*	*55.6%*	*P2*
gemma-4-31b-it	*C3*	*11.1%*	*66.7%*	*P2*
mistral-medium-3.5-128b	C1	66.7%	66.7%	equal
mistral-medium-3.5-128b	C2	33.3%	33.3%	equal
mistral-medium-3.5-128b	C3	11.1%	33.3%	P2

Measure (lower = better unless stated)	P1 – all 12	P2 – all 12	Better (all 12)	P1 – without Gemma (9)	P2 – without Gemma (9)	Better (without Gemma)
Mean yes rate (macro-average)	32.4%	42.6%	**P1**	34.6%	39.5%	**P1**
Median yes rate	33.3%	33.3%	**≈ equal**	33.3%	33.3%	**≈ equal**

Worst cell (highest yes rate)	66.7%	66.7%	**≈ equal**	66.7%	66.7%	**≈ equal**
Cells with ≤ 1 of 9 hallucinated (higher = better)	3	1	**P1**	2	1	**P1**
Cells where this style hallucinates more (head-to-head)	2	6	**P1**	2	4	**P1**
Count "P1" better			**3**			**3**
Count "P2" better			**0**			**0**
Count "≈ equal"			**2**			**2**

Reading (hallucinations): across all 12 pairs, P1 is better on 3 of 5 measures, P2 on 0, 2 about equal. Mean yes rate P1 32.4% vs P2 42.6%; P2 hallucinates more in 6 cells, P1 in 2.
Overall (5.1 + 5.2): on faithfulness P1 is at least as good as P2 (better on 3 of 7 measures, P2 on 1); on hallucinations P1 is better on 3 of 5, P2 on 0. The strategy-level data favours P1; the single best cell (GPT-OSS × C2 × P2) is the exception, not the pattern.

5.3 Practicality per model × configuration (tie-breaker)

Duration = median ms of the 9 explanations per cell (median because of outliers, e.g. DeepSeek C2 P1). Words = mean words per explanation. "≈ equal" = values differ by < 5 %.

Model	Config	Median ms P1	Median ms P2	Faster	Mean words P1	Mean words P2
deepseek-v4-flash-0731	C1	1,264	1,736	P1	61.8	70.1
deepseek-v4-flash-0731	C2	87,962	8,238	P2	64.7	77.8
deepseek-v4-flash-0731	C3	1,749	3,349	P1	54.3	62.6
openai-gpt-oss-120b	C1	4,323	4,314	≈ equal	56.1	59.4
openai-gpt-oss-120b	C2	11,755	18,007	P1	55.0	59.7
openai-gpt-oss-120b	C3	5,410	6,309	P1	51.0	48.1
gemma-4-31b-it	*C1*	*1,068*	*1,909*	*P1*	*35.0*	*67.6*
gemma-4-31b-it	*C2*	*5,826*	*6,547*	*P1*	*36.7*	*42.4*
gemma-4-31b-it	*C3*	*2,850*	*4,197*	*P1*	*38.1*	*52.4*
mistral-medium-3.5-128b	C1	1,583	2,329	P1	47.3	56.0
mistral-medium-3.5-128b	C2	5,514	7,166	P1	43.6	51.1
mistral-medium-3.5-128b	C3	2,810	3,817	P1	43.4	55.8

Measure (lower = better unless stated)	P1 – all 12	P2 – all 12	Better (all 12)	P1 – without Gemma (9)	P2 – without Gemma (9)	Better (without Gemma)
Mean duration (mean of cell medians, ms)	11,010	5,660	**P2**	13,597	6,141	**P2**
Median duration (median of cell medians, ms)	3,587	4,256	**P1**	4,323	4,314	**≈ equal**
Slowest cell (ms)	87,962	18,007	**P2**	87,962	18,007	**P2**
Cells where this style is faster (higher = better)	10	1	**P1**	7	1	**P1**
Mean words per explanation	48.9	58.6	**P1**	53.0	60.1	**P1**
Count "P1" better			**3**			**2**
Count "P2" better			**2**			**2**
Count "≈ equal"			**0**			**1**

Reading (practicality): P1 is faster in 10 of 12 cells, P2 in 1. P2 wins mean duration and slowest cell only because of one outlier cell (slowest P1 cell: 88.0 s). P1 explanations are shorter (48.9 vs 58.6 words).

5.4 Overall result: which prompting strategy wins?

Decision rule: faithfulness (5.1) and hallucinations (5.2) decide; practicality (5.3) is only used as a tie-breaker. A category is won by the style that is better on more measures.

Category	P1 better – all 12	P2 better – all 12	≈ equal – all 12	Winner – all 12	P1 better – without Gemma (9)	P2 better – without Gemma (9)	≈ equal – without Gemma (9)	Winner – without Gemma (9)
5.1 Faithfulness (7 measures)	3	1	3	**P1**	5	1	1	**P1**
5.2 Hallucinations (5 measures)	3	0	2	**P1**	3	0	2	**P1**
5.3 Practicality – tie-breaker (5 measures)	3	2	0	**P1**	2	2	1	**tie**
OVERALL WINNER				**P1**				**P1**

Reading: overall winner across all 12 pairs: P1; without Gemma (9 pairs): P1. Faithfulness: P1 3 vs P2 1 measures; hallucinations: P1 3 vs P2 0. The single best cell (GPT-OSS × C2 × P2) is the exception, not the pattern.